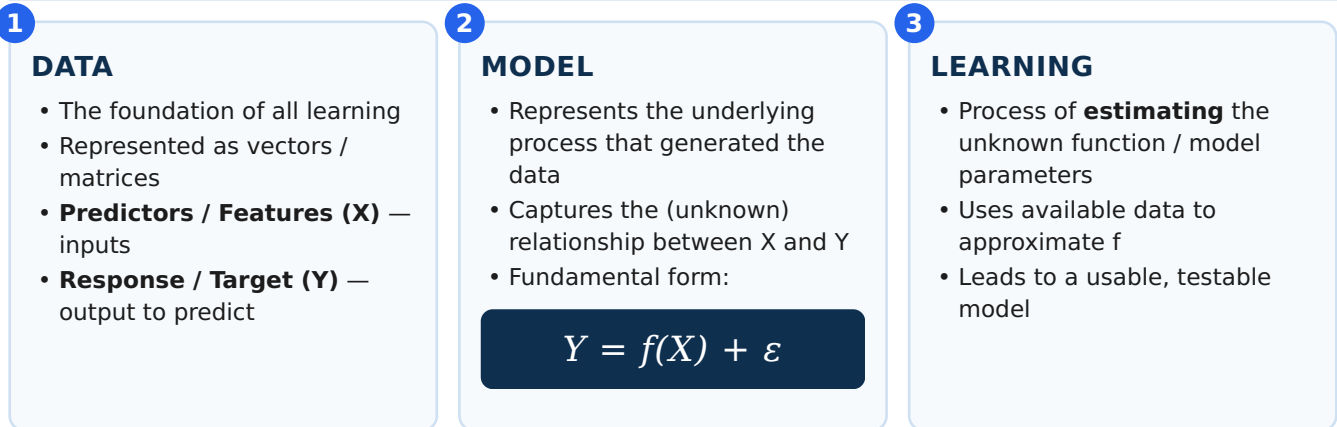


Lecture 1 — Introduction to Statistical & Machine Learning

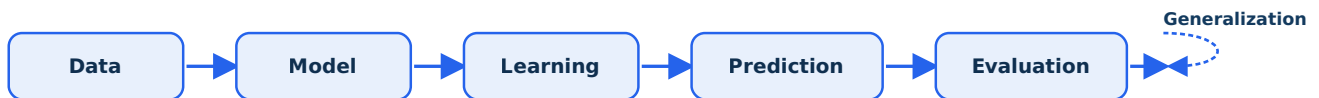
Exam & Interview
Revision Sheet

Statistical & Machine Learning: Methods and algorithms that extract useful information or patterns from data.

The Three Interconnected Pillars



The Learning Pipeline



REAL-WORLD EXAMPLE

Predicting **house price** using area, number of bedrooms, location and age. Here, area/bedrooms/location/age are **predictors (X)**, and house price is the **response (Y)**.

Data as a Vector / Matrix — Illustration

Each row is one observation. Predictor columns together form matrix X ; the last column forms response vector Y .

Area (sq.ft)	Bedrooms	Location	Age (yrs)	Price (Y)
1200	2	Suburb	5	₹55 L
1800	3	City Center	2	₹95 L
950	2	Suburb	12	₹38 L
2400	4	City Center	1	₹1.4 Cr

Columns 1-4 → predictor matrix X ($n \times p$) | Column 5 → response vector Y ($n \times 1$) | n = observations, p = predictors.

Mathematical Foundation & Error

The Statistical Learning Model

$$Y = f(X) + \varepsilon$$

Symbol	Meaning
Y	Quantitative response / target variable
$X = (X_1, X_2, \dots, X_p)$	Set of p predictors / features
f	Unknown, fixed underlying function relating X to Y
ε	Random error term (noise), independent of X , with $E[\varepsilon] = 0$

KEY IDEA

Learning means estimating f from data. Once estimated ($\hat{f}$), we can predict Y for new X using $\hat{Y} = \hat{f}(X)$.

Decomposing Prediction Error

The expected squared prediction error can be written as:

$$E[(Y - \hat{Y})^2] = [f(X) - \hat{f}(X)]^2 + \text{Var}(\varepsilon)$$

Prediction Error = **Reducible Error** + **Irreducible Error**

REDUCIBLE ERROR

Comes from an imperfect estimate $\hat{f}$. Can potentially be reduced through:

- Better model choice
- Better / more relevant features
- Better / more data
- Better parameter estimation

IRREDUCIBLE ERROR

Comes from $\text{Var}(\varepsilon)$ — inherent randomness. Cannot be eliminated, even with a perfect model:

- Unmeasured variables
- Natural variability in outcomes
- Inherent noise in the process

REAL-WORLD EXAMPLE

Two houses with nearly identical characteristics can still sell for different prices due to negotiation, seller urgency, market fluctuation, and other unobserved factors — this variability is **irreducible error**.

Generalization

KEY IDEA

A good model should perform well not only on **training data** but also on **unseen data**.

Overfitted / Memorized Model



High training fit · Poor on unseen data

Generalized Model



Strong on training AND unseen data

Statistics vs Machine Learning & Key Takeaways

Questions Statistical Learning Tries to Answer

- What are the predictors and the response?
- Is there a relationship between X and Y?
- Is the relationship linear or nonlinear?
- What is the underlying data-generating process?
- Is the available data sufficient for inference?
- Can the model make useful predictions?

Predictor Selection

Not every available variable is a **useful predictor**. Predictor selection means identifying variables that genuinely relate to the response.

REAL-WORLD EXAMPLE — SALARY PREDICTION

Useful predictors: **Likely irrelevant:**

- Experience
- Education
- Technical skills
- Favorite color

EXAM POINT

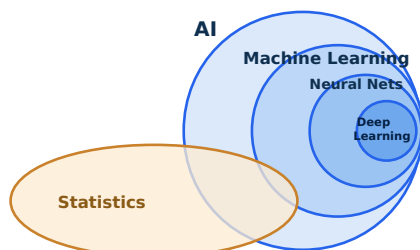
Correlation with the response does **not** automatically imply causation.

Statistics vs Machine Learning

Aspect	Statistics	Machine Learning
Primary emphasis	Inference / explanation	Prediction / generalization
Typical concern	Interpretability	Predictive performance
Model complexity	Often simpler	Can be highly complex
Parameters	Often fewer	Can be many
Main question	Why / how?	Can we predict?

*Note: This is a **general tendency**, not an **absolute rule** — modern statistics and ML overlap heavily.*

AI, ML & Deep Learning — Nested View



Examples from the lecture:

- Linear regression
- Random forests
- Boosting
- Kernel methods
- Neural networks
- Deep learning

ONE-MINUTE REVISION — Remember These

1. ML extracts useful information / patterns from data.
2. Data, Model, and Learning are interconnected.
3. $Y = f(X) + \varepsilon$ is the fundamental statistical-learning formulation.
4. X represents predictors; Y represents the response.
5. f is the unknown underlying relationship.
6. ε represents random error / noise.
7. Learning estimates the unknown function / model parameters.
8. Prediction uses $\hat{Y} = \hat{f}(X)$.
9. Prediction error = Reducible Error + Irreducible Error.
10. A useful model must generalize to unseen data.